

7 The curve C has equation $y = \frac{2x^2 - 5x}{2x^2 - 7x - 4}$.

(a) Find the equations of the asymptotes of C .

[2]

[illegible]

(b) Find the coordinates of any stationary points on C .

[4]

[illegible]



(c) Sketch C , stating the coordinates of the intersections with the axes.

[3]

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(d) Sketch the curve with equation $y = \left| \frac{2x^2 - 5x}{2x^2 - 7x - 4} \right|$.

[1]



(e) Find in exact form the set of values of x for which $\left| \frac{2x^2 - 5x}{2x^2 - 7x - 4} \right| < \frac{1}{9}$. [5]

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